



EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 32-302-1
Digital Earth Resistance Tester	27-11-2024

EDMS 32 - 302 - 1

SPECIFICATION
FOR
DIGITAL EARTH RESISTANCE TESTER

Issue: Nov. -2024/ Rev- 1

- This revision contains option items that must be selected by EDC before bidding.

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1. SCOPE

This specification covers the minimum technical requirements for supply, inspection and delivery of portable digital instrument to measure the earth resistance and/or soil resistivity of power distribution networks. The required tester is designed to perform high accuracy measurements. The required tester is intended to operate in both indoor and outdoor of the power networks of Egyptian Electricity Distribution Companies (EDCs).

2. APPLICABLE STANDARDS

The equipment/material covered in this specification shall comply with the latest edition/amendment of the standards given in Table (1). Where any provision of this specification differs from those of the standards listed below, the provisions of this specification shall apply. Any deviations from the listed standards or the provisions of this specification shall be clearly set out in the offer.

Table (1)

Standard No.	Description
IEC 61010-1	Safety requirements for electrical equipment for measurement, control, and laboratory use - Part 1: General requirements
IEC 61010-2-030	Safety requirements for electrical equipment for measurement, control, and laboratory use - Part 2-030: Particular requirements for equipment having testing or measuring circuits
IEC 61010-031	Safety requirements for electrical equipment for measurement, control, and laboratory use - Part 031: Safety requirements for hand-held and hand-manipulated probe assemblies for electrical test and measurement

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IEC 61557-1	Electrical safety in low voltage distribution systems up to 1000 V AC and 1500 V DC - Equipment for testing, measuring or monitoring of protective measures - Part 1: General requirements
IEC 61557-5	Electrical safety in low voltage distribution systems up to 1000 V AC and 1500 V DC - Equipment for testing, measuring or monitoring of protective measures - Part 5: Resistance to earth
EN 50581	Technical documentation for the assessment of electrical and electronic products with respect to the restriction of hazardous substances
IEC 61326-1	Electrical equipment for measurement, control and laboratory use - EMC requirements - Part 1: General requirements
IEC 61326-2-2	Electrical equipment for measurement, control and laboratory use - EMC requirements - Part 2-2: Particular requirements - Test configurations, operational conditions and performance criteria for portable testing, measuring and monitoring equipment used in low-voltage distribution systems
IEC 60529	Degrees of Protection Provided by Enclosures (IP Code)

3. ENVIRONMENTAL CONDITIONS

The behavior of the required tester shall be guaranteed under the environmental conditions given in Table (2).

Table (2)

Operating temperature	-10°C to +50°C
Storage temperature	-10°C to +50°C
Maximum relative humidity	80 %
Altitude	Up to 1000 m above mean sea-level

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4. MAIN FEATURES

The digital earth resistance tester should have the following main features:

- Robust, compact and portable instrument designed for measuring electrode resistance with user-friendly operation in a complete safety.
- The instrument should have facility to measure Ground Resistance by standard Fall of Potential method using 3 terminals method by Current and Potential spikes in accordance to IEC 61557-5. The equipment must also enable to measure the 2 terminals measurement. (4 terminals is optional in case of soil resistivity according to __EDC requirements).
- The instrument also includes a voltmeter to measure the ground voltage.
- Large LCD display with backlight. The backlight should facilitate working at a dimly illuminated location or at nighttime work.
- Indicate operation modes and monitor correct/non-correct measurements.
- The tester should automatically power off / saving if no button is pressed for 10 minutes. Backlight should automatically turns off if there is no activity for 2 minutes.
- Hold function to hold the measured value.
- Maximum Manufacturing date of one year before delivery date.

5. TECHNICAL PARAMETERS

The digital earth resistance tester should have the following requirements:

- Earth resistance reading range: from 0 to 2000 Ω in auto-range. Resistance Accuracy: $\pm 2\%$ of the measured value or better.
- Resolution: $\leq 0.1 \Omega$.
- Display: at least 4 digits to display measurements and warning messages.

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- Test frequency: 128 Hz $\pm$ 2 Hz.
- Earth voltage reading range: from 0 to 300 V AC (45-65 Hz) / $\pm$ 0 to $\pm$ 300 V DC.
Accuracy: $\pm$ 1% of the measured value or better.
- To allow accurate testing in noisy environments, the instrument should be capable of rejecting noise. Noise rejection up to 25 V rms.
- Equipped with LCD display viewable in bright sunlight and low-light levels.
- Dust and water proof construction (IP 54 or better) in accordance to IEC 60529
- Power source:
 - Battery operation $\geq$ 400 measurement of continuous operation.
 - Rechargeable battery available in the local markets. (suitable charger +one set of spare battery should be delivered)
- The Instrument should automatically check and display the battery status Indication.
- Standard software for data recovery and processing of the results.
- Memory to store more than 512 record. (FIFO First in First out)
- The portable tester shall be light in weight not more than 1.5 kg without accessories.

6. ACCESSORIES

Each digital earth resistance tester should be attached with the following accessories:

- Complete cable set of earth resistance test leads (30 m, 15 m and 5 m of different colors) (in case of soil resistivity one more cable of 5 m must be delivered)
- Measurement probes.
- Complete set of auxiliary earth spikes.
- Carrying case for instrument and all supplied accessories.

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7. MARKING

- The tester shall be marked legibly and indelibly with the following information:
 - Name or trade mark of the manufacturer
 - Country of origin
 - Type/model serial number
- All marking shall to be written in English with suitable font.

8. SUBMITTALS AND OTHER REQUIREMENTS

- Tenderer must submit with the offer original catalogue containing full technical data and the instruction manual of operation and maintenance.
- The supplier should provide a device sample and furnish a seminar about the device capabilities (upon request)
- The supplier should provide test certificates, recent calibration results (not older than 3 months before delivery date) and Origin country certificate.

9. GUARANTEE

- The supplier guarantee the supplied instruments against any defects arising out of faulty design or workmanship, or of defective material for a period of two (2) years from the date of delivery.

10. TRAINING and CUSTOMER SUPPORT

- The supplier shall provide adequate on-site training for a sufficient number of ____ EDC representatives, to be agreed by (.....EDC) and supplier.
- The Supplier must have a dedicated service center and support line in Egypt to cover any maintenance or repair requirements.

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- __ EDC reserves the right to claim a free after-sales service for the entire warranty period. After-sales service should include and not limited to the following:
 - Technical support.
 - Maintenance.
 - Original Spare parts.

11. TECHNICAL DATA SCHEDULE

- The tenderer must fill in thoroughly the attached technical data schedule.
- Any offer does not accompanied with clear and complete technical data schedule shall be rejected.

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TECHNICAL DATA SCHEDULE DIGITAL EARTH RESISTANCE TESTER

__ EDC Inquiry No: _____ Item No: _____

No.	DESCRIPTION	UNIT	__ EDC Specified Values	Vendor Proposed Values
1	Manufacturer's name			
2	Country of Origin			
3	Tester Model			
4	Digital measuring range	Ω		
5	Accuracy of measuring range	%		
6	Resolution	Ω		
7	Test frequency	Hz		
8	Battery capacity	A,h		
9	Battery voltage	V		
10	Battery is rechargeable?			
11	Guarantee	Month	24	
12	Accessories meet all the requirements?		Yes	
13	Training and customer support meets all the requirements?		Yes	
14	Submittals meet all the requirements?		Yes	

I / We guarantee the technical data given above for the offered material/equipment.

Description	Manufacturer of Material/Equipment	Vendor/Supplier
Vendor/Supplier		
Location and Office Address		
Name and Signature of Authorized Representative with Date		
Official Seal / Stamp		